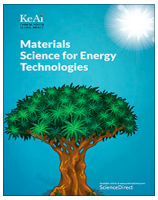
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# Chemical modification of sunflower waste cooking oil for biolubricant production through epoxidation reaction



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ABSTRACT

There are increasing concerns and growing regulations over contamination and environmental pollution. One of the major concerns is the pollution caused by mineral oils. As the world oil reserves are dwindling the pressures for finding alternative replacements are increasing. Vegetable oils are a biodegradable and renewable source of lubricants. Thus, they seem to be attractive candidates for the replacement of mineral oils. In this study, chemical modification of sunflower waste cooking oil through epoxidation reaction was used to improve its physicochemical properties to be used as biolubricant. Then, using the response surface method (RSM), the reaction yield was statistically modeled as a quadratic regression equation and optimal points were obtained for independent variables. The maximum amount of reaction yield was 82.9% and optimum values for independent variables were; 53.7 g of acetic acid, 30.1 g of $H_2O_2$, and 4.1 h of reaction time. The obtained optimum epoxide mixture was used as the starting material to produce SWCO triester during three chemical reactions and analyzed with the help of FTIR spectrum analysis. It was found that the physicochemical properties of the final product (SWCO triester) met the requirements of the ISO VG10 standard.

## 1. Introduction

Today, the decrease in oil reserves and also increase in energy price, economic consumption of energy, and its optimization in energy consumptive systems have become an indispensable task. Hence, scientist's attention to renewable energy sources such as wind, solar, and biofuels such as biodiesel and bioethanol has increased [1]. Internal combustion engines employ the largest share of oil consumption and are one of the biggest consumers of energy. Thus, considering ways to reduce energy consumption in these engines are of utmost importance. Since one-third of energy consumption in internal combustion engines is due to frictional losses, it can be concluded that lubricating the necessary parts of these engines to reduce abrasion and friction of moving parts can have a tremendous effect on energy and fuel consumption [2]. Generally, engine oil is one of the vital fluids in the engine and by lubricating different parts, it plays the role of blood in the body's blooding system. On the other side, like the other oils utilized in the engine, this fluid is petroleum-based and it depends on expensive and non-renewable sources, meaning that one cannot rely on them for a long time. Furthermore, millions of tons of lubricants are used annually and they are disposed to nature as burned oil, causing environmental pollution [3]. This is besides the damage which is inflicted by burning the lubricant with engine fuel that makes air pollution. Hence, nowadays lubricants are made from renewable and eco-friendly resources like vegetable oils. The vegetable-based lubricants are called biolubricants and they depend on renewable vegetable resources. Due to their good biodegradation, they do not pollute the environment [4,5].

Most of the vegetable oils are made from Triacylglycerol. Triacylglycerol is a glycerol triester in which three fatty acids are linked through an ester linkage to three glycerol hydroxyl (Fig. 1). Physical and chemical features of vegetable oils mostly depend on the composition and distribution of fatty acids and glycerol structure of Triacylglycerol [6]. Triacylglycerol's structure is suitable for lubrication since the combination of long-chain fatty acids with carboxyl group engenders very strong lubrication. Most fatty acids of vegetable oils have 14 to 22 carbon atoms with different saturations. Triacylglycerol molecules in comparison to hydrocarbons delineate better interaction with metal surfaces and this makes them a very strong lubricant on the metal surface. Hence, they are very useful in reducing abrasion and friction [7].

But vegetable oils cannot be used as lubricants directly. According to Fig. 1, double bonds in fatty acids of triacylglycerol

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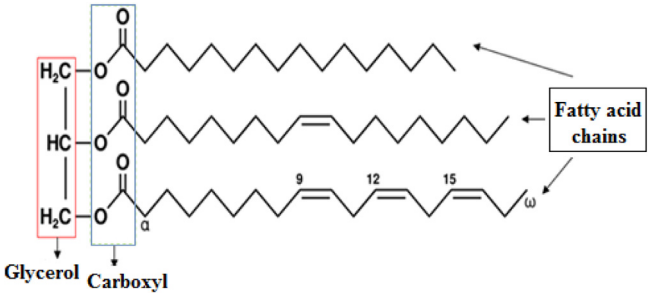



**Fig. 1.** The vegetable oils' Triacylglycerol structure.

are reactive and they react with oxygen in the air, hence they cause oxidation and finally engender deposition of insoluble materials, increase in acid value, and oil viscosity. Also, $\beta$ hydrogen in the glycerol section can be easily removed from molecule structure and leads ester to converted into acid and olefin. Disadvantages of direct applications of vegetable oils as lubricants are low stability against oxidation and low pour point (tend to freeze at low temperatures and poor flow behavior) [8,9]. Researchers have employed various methods to analyze the possibility of using vegetable oils as lubricants.

Some of the researchers studied replacing polyols such as Trimethylolpropane with glycerol structure of vegetable oils to increase oxidation stability [10,11]. In another method, epoxidation was used to stabilize against canola oxidation [12,13]. Genetic change, hydrogenation, and transesterification reaction of vegetable oil with long-chain alcohols (with 8–15 Carbon atoms) were also studied [2,14,15]. Epoxidation is one of the main additive reactions to double bonds that lead to the formation of oxirane rings and hence increases the oxidation stability of vegetable oils [16,17]. This reaction is done in presence of proxy acids (like performic acid and peracetic acid) and hydrogen peroxide and the result of this experiment is a peracid that epoxidates double bond (Fig. 2). Few studies have investigated the optimization of epoxidation reaction and until now there has been a lack of firm understanding and research into how the interaction between the experimental factors will affect the epoxide yield which will be addressed in this paper.

## 2. Materials and method

### 2.1. Materials

In this study, sunflower waste cooking oil (SWCO) was selected as base vegetable oil with a high percentage of oleic fatty acid as its main fatty acid. The reason for choosing SWCO was its lubrication effect, low melting point (-10 to –22°C), and availability. The physicochemical properties of SWCO and the values of its fatty acids are demonstrated respectively in Table 1 and Table 2.

Reaction materials were as follows: A mixture of oleic acid provided from Aland chemical industry, Hydrogen peroxide 30%, Acetic acid, Sulfuric acid 99%, Stearic acid 99%, Palmitic acid 99%, para toluene sulfonic acid 98%, toluene 98.2%, Normal Hexane 99%, Potassium hydroxide 99%, Methanol 99.7%, Oleic acid 99%, Normal Heptane 99.9%, Isopropyl alcohol 99%, Magnesium sulfate 97%, Formic acid 88%, Normal Propanol, Methyl Orange, Potassium iodide, Hydrochloric acid. The devices used in this study include: Rotary evaporator model EKIRTS 202, Viscometer RaapnotnA model MVS-3000, Spectroscope FTIR model RekurB, Flashpoint analyzer (Analyze-p611, Belgium), pour point and cloud point determination with alcohol bath device model AETS made in England and Perkin Elmer Gas Chromatography (GC) model Clarus580 made in the United States. Extraction and derivation of fatty acids in SWCO was performed according to Metcalf method [18]. In this method, methyl fatty acids were obtained using boron trifluoride and the percentage of fatty acid methyl esters, which is equivalent to the fatty acids in the oil, was obtained through GC [19].

### 2.2. Research method

#### 2.2.1. Epoxidation reaction

To make epoxide, 20 g of fatty acid was poured into a 250 ml flask and it was placed on the Styrene heater to reach its temperature of 65°C. Then, 50 ml of acetic acid or formic acid 88% was added. When the mixture reached a constant temperature, a burette was used to add 11.30 ml of Hydrogen peroxide in one hour. Since the oxidation reaction performs better in acetic condition, the acetic acid was used as the catalyst. After adding hydrogen peroxide, it was mixed with a magnetic mixer for 5 h at 500–600 rpm. The color of the mixture turns to light yellow. In the next step, the mixture was poured into the separation funnel and after separation

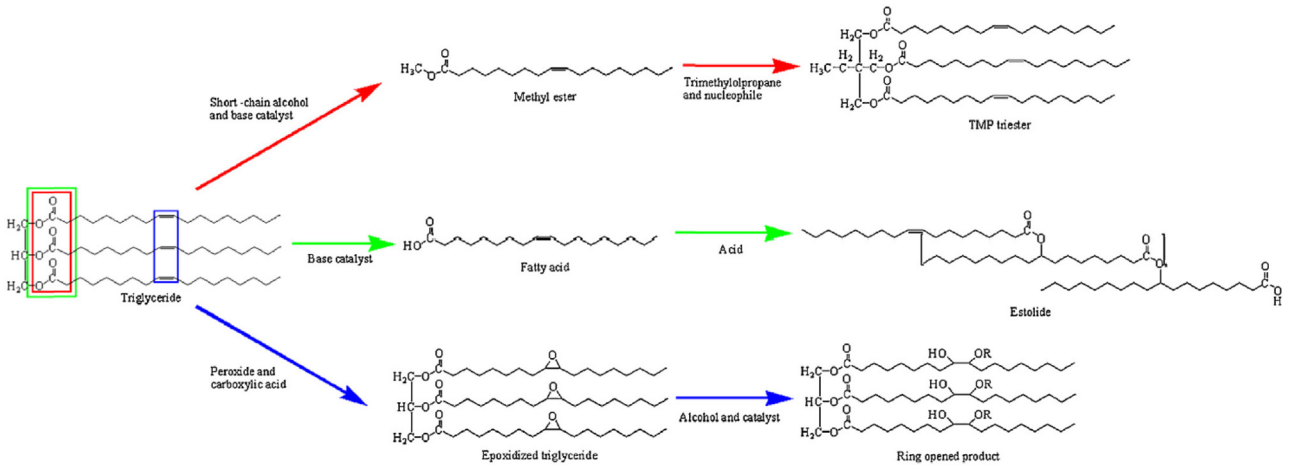



**Fig. 2.** Reaction pathways of the three main chemical modification methods. Red, green, and blue solid arrows represent the pathway of esterification/transesterification, estolide formation, and epoxidation/ring-opening, respectively [12]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 1**
Physicochemical properties of SWCO.

| Visual | Yellow | Color |
|---|---|---|
| ASTM-D664 | 2.31 | Free acidity (according to oleic acid) |
| ASTM-D1298 | 0.905–0.91 | Density |
| ASTM-D94 | 189–194 | Saponification value |
| EN 14,111 | 110–145 | Iodine value |
| – | 70–80% | Unsaturated fatty acids |

**Table 2**
Waste cooking oil fatty acids and their value.

| Value | Fatty acid |
|---|---|
| 1.37 | Myristic acid (C 14:0) (wt%) |
| 6.66 | Palmitic acid (C 16:0) (wt%) |
| 3.04 | Stearic acid (C 18:0) (wt%) |
| 17.87 | Oleic acid (C 18:0) (wt%) |
| 72.13 | Linoleic acid (C 18:2) (wt%) |
| 854.139 | Average molar weight of waste cooking oil (g/mole) |




**Fig. 3.** Purifying the epoxide of waste cooking oil by water washing.

into 2 phases, the bottom phase was removed (Fig. 3). To separate the catalyst (acetic acid), the remaining phase was washed with distilled water.

To optimize the reaction of producing epoxide, Design-Expert 7.0 software and the Box-Behnken method were employed. The number of experiments was 20 and among them, 3 were dedicated to the central point. Table 2 shows the measured parameters of the experiment. To determine the percentage of epoxy, 1.5 g of epoxide was poured into a 100 ml balloon and 25 ml normal propanol was added and it was refluxed while steering.

After dissolution, a few drops of methyl orange and newly generated potassium iodide (3 g KI in 5 ml of distilled water) were added and it was refluxed again. Finally, the mixture was titrated with normal HCL get to the finished yellow state of stability and epoxide value was obtained with Eq. (1):

$$Y = \frac{\mathbf{HCLnormality} \times \mathbf{HCLconsumption}(\mathbf{ml}) \times 100}{\mathbf{Sampleweight}(\mathbf{gr})} \quad (1)$$

In which, Y is the epoxide reaction efficiency (dependent variable).

*2.2.2. Synthesis of monoester from epoxide*

To perform this reaction, 31 g of epoxide with 6 g palmitic acid and 5 g P-toluene sulfonic acid was dissolved in 70 ml toluene and was poured into a two-necked flask. The reaction was performed under heat and stirring at 70°C for 90 min. Then the temperature was increased to and the mixture was stirred and refluxed at 90°C for 90 min. After the completion of the reaction, the content of the flask was poured into the funnel and was kept in it for 12 h at ambient temperature. Then the lower phase was removed and the upper phase was washed 5 times with 50 ml of distilled water. The acaline phase was dehydrated by magnesium sulfate and toluene was removed from the reaction mixture using the vacuum rotary apparatus at the speed of 100 rpm and 90°C.

*2.2.3. Synthesis of diester from monoester*

10 g of monoester with 27.8 g of octanol were poured into a two-necked flask and 4 drops of sulfuric acid (10%) was added as a catalyst. The reaction was carried out at 60°C with stirring for 10 h. After the reaction completion, the content of the flask was poured into the funnel. To separate and purify the resulting product, 50 ml of n-hexane were added and after separating the aqueous phase from the organic, 5 ml of saturated sodium bicarbonate was added to the organic layer, and the lower phase was removed.

*2.2.4. Synthesis of triester from diester*

14 g of diester obtained from the previous stage was weighted and with 14 g stearic acid and a few drops of sulfuric acid (10 mol%) was added as a catalyst inside the two-necked. The mixture was refluxed for 10 h at 80°C temperature. After the reaction completion, the content of the flask was poured into the funnel. To separate and purify the obtained product, 60 ml of n-hexane was added and then 30 ml of saturated sodium hydrogen carbonate was added and the low phase was removed, and then 30 ml of saturated brine was added. Then the organic and aqueous phases were separated and the pH at this stage reached 7. Finally, after removing the n-hexane from the organic phase by distillation with anhydrous magnesium sulfate, the product was dewatered and cleared up by vacuum until the triester was obtained. Fig. 4 indicates the comparison of results between experimental and model data.

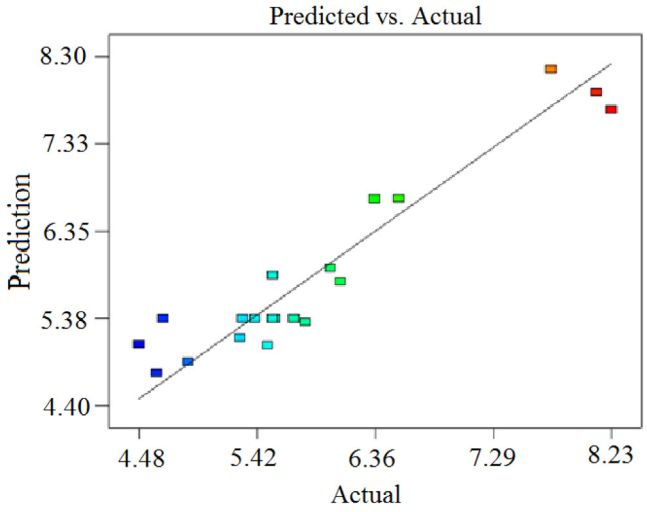



**Fig. 4.** Comparison between experimental data and model results.

## 3. Results and discussion

### 3.1. Epoxidation process modeling

Table 3 indicates the results of variance analysis related to quadratic regression for various variables. According to Table 4, and removing the parameters which are inefficient in obtaining results, Eq. (2) was obtained.

$$R = 7.14650 + 0.057581 \times A - 0.047197 \times B - 0.70377 \times C - 1.00000 \times AB + 5.5192 \times AC + 2.40380 \times BC + 1.86631 \times A2 - 3.46962 \times B2 + 0.020358 \times C2 \quad (2)$$

In this equation, R is the response variable, A is the measure of acetic acid, B is the measure of hydrogen peroxide and C indicates the reaction time. The plus sign (+) in front of each section indicates the synergistic effect and the minus sign (−) signifies the negative impact of parameters on the response.

### 3.2. Epoxidation process optimization

To optimize the epoxide production process, the conversion percentage is set to maximum and utilized hydrogen peroxide is set to minimum. To this end, according to Table 5, boundary conditions of independent variables and object variables were determined. Due to the equal importance between independent variables in weighting the independent variables of the objective function, weight was determined as 1. Table 6 shows the optimized reaction points for maximum efficiency and due to boundary conditions and applied to weigh, Acetic acid 53.66 g, Hydrogen Peroxide 30.11 g and time 4.09 h were determined by the software.

With the proximity of experimental results of optimized values along with model results, the validity of optimized values was obtained. Also, the diagram of Figs. 5–7 delineates the changes of the response variable (epoxide production efficiency) to the independent variables of acetic acid, hydrogen peroxide value, and time of the reaction are shown in three-dimensional diagrams and with considering the simultaneous changes of two independent variables respectively.

Fig. 5**-a** shows the effect of acetic acid and hydrogen peroxide on the epoxide amount, where the temperature is constant. As can be seen, an increase in acetic acid value has a stronger effect on epoxide amount than an increase in $H_2O_2$ value. Because acetic acid is a better transferor for oxygen. The obtained results are comparable with Chen et al., 2019 [20]. In Fig. 5**-b**, the effects of reaction time and hydrogen peroxide values on the epoxide amount have been illustrated. It can be understood that an increase in reaction time increases the probability of breaking in epoxies returning them to their initial state and therefore decreases the epoxide amount. Whereas an increase in $H_2O_2$ slightly increases the epoxide amount. As can be seen, the highest epoxide value was obtained 4 h and 30 ml. According to Fig. 5**-c**, it can be seen that increase in acetic acid value increases the amount of epoxide, and an increase in reaction time decreases the epoxide amount. Thereupon, the value of time should not be less than a certain quantity, otherwise, the epoxidation could not be fulfilled.

### 3.3. Analysis of triester synthesis process using FTIR spectroscopy

Fourier Transform Infrared Spectroscopy, also known as FTIR Analysis or FTIR Spectroscopy, is an analytical technique used to identify organic, polymeric, and in some cases, inorganic materials. The FTIR analysis method uses infrared light to scan test samples and observe chemical properties. Figs. 6–9 show the FTIR analysis of epoxide, monoester, diester, and triester obtained from the experiments. As is illustrated in Fig. 6, the peaks in 1844 $cm^{-1}$ wavenumber at epoxide FTIR spectra are related to the carbons in oxirane ring, and peaks in 2925 $cm^{-1}$ and 12854 $cm^{-1}$ are resulted by stretching vibration of $CH_2$ (aliphatic carbons). Also, the peaks in 1743 $cm^{-1}$ and 1465 $cm^{-1}$ are related to the stretching vibration of carbonyl and bending vibration of $CH_2$ respectively. Furthermore, the peaks in 723 $cm^{-1}$, 1378 $cm^{-1,}$ and 3529 $cm^{-1}$ are results of $CH_2$ oscillatory vibration and O–H bonds' bending and stretching vibrations respectively.

In the FTIR spectra for monoester (Fig. 7), the peaks in 2919 $cm^{-1}$ and 2850 $cm^{-1}$ wavenumbers, belong to $CH_2$ stretching vibration. The peaks in 1706 $cm^{-1}$ and 2850 $cm^{-1}$ wavenumbers are related to the stretching vibration of carbonyl ester group and carboxylic acid which represent the oxyran ring opening.

In the diester spectrum, the peak in 3350 $cm^{-1}$ shows the stretching vibration of the OH bond, and the peaks in 2925 $cm^{-1}$ and 2855 $cm^{-1}$ indicate the stretching vibration of $CH_2$. The peak in 1737 $cm^{-1}$ is related to stretching vibration of carbonyl and the peaks in 1117 $cm^{-1}$, 1174 $cm^{-1,}$ and 1246 $cm^{-1}$ wavenumbers are related to stretching vibration in the C-O band in ester. Also, the peaks 1465 $cm^{-1}$, 1377 $cm^{-1,}$ and 722 $cm^{-1}$ belong to bending vibrations in $CH_2$ and $CH_3$ and oscillatory vibration in $CH_2$ respectively (Fig. 8).

**Table 3**
The measured parameters of the experiment according to RSM.

| (%)Conversion | (h) Time | (ml) Acetic acid | (ml) $H_2O_2$ | Experiment number |
|---|---|---|---|---|
| 48.7 | 15 | 30 | 30 | 1 |
| 44.8 | 10.30 | 50 | 90 | 2 |
| 46.7 | 10,30 | 50 | 50 | 3 |
| 63.5 | 6 | 30 | 30 | 4 |
| 77.5 | 6 | 70 | 30 | 5 |
| 58 | 15 | 30 | 70 | 6 |
| 55.4 | 10.30 | 50 | 50 | 7 |
| 52.8 | 6 | 70 | 70 | 8 |
| 82.3 | 10.30 | 50 | 10 | 9 |
| 60.8 | 6 | 30 | 70 | 10 |
| 46.2 | 10.30 | 10 | 50 | 11 |
| 35 | 10.30 | 50 | 50 | 12 |
| 45 | 10.30 | 50 | 50 | 13 |
| 65.4 | 15 | 70 | 30 | 14 |
| 57.1 | 10.30 | 50 | 50 | 15 |
| 55.6 | 10.30 | 50 | 50 | 16 |
| 81.1 | 10.30 | 50 | 50 | 17 |
| 55 | 15 | 70 | 70 | 18 |
| 60 | 10.30 | 90 | 50 | 19 |
| 55.4 | 19.30 | 50 | 50 | 20 |

**Table 4**
Variance analysis (Quadratic regression model).

| Df | MS | F-value | P-value | SS | Source |
|---|---|---|---|---|---|
| 9 | 2.21 | 9.52 | 0.0008 | 19.85 | Model |
| 1 | 1.01 | 4.35 | 0.0636 | 1.01 | Acetic acid |
| 1 | 4.54 | 19.59 | 0.0013 | 4.54 | $H_2O_2$ |
| 1 | 5.78 | 93.24 | 0.0005 | 5.78 | Time |
| 1 | 2.17 | 9.38 | 0.0120 | 2.17 | Acetic acid× $H_2O_2$ |
| 1 | 0.071 | 0.31 | 0.5929 | 0.071 | Acetic acid× Time |
| 1 | 0.93 | 4.03 | 0.0726 | 0.93 | Time × $H_2O_2$ |
| 10 | 0.23 | | | 2.32 | Residual |
| 5 | 0.33 | 2.43 | 0.1763 | 1.64 | Incompatibility |
| 5 | 0.14 | | | 0.68 | Error |
| 19 | | | | 22.17 | Total |

**Table 5**
The boundary condition of independent and target variables.

| Target | Lower limit | Upper limit | Weight (g) | Variable |
|---|---|---|---|---|
| 20–70 | 20 | 70 | 1 | Acetic acid |
| 30–70 | 30 | 70 | 1 | $H_2O_2$ |
| 4–15 | 4 | 15 | 1 | Time (h) |
| 40.48–80.23 | 40.48 | 80.23 | 1 | Conversion (%) |

**Table 6**
Experimental results of epoxide conversion based on RSM suggestions for each parameter.

| Favorability | Epoxide yield (%) | Time (h) | $H_2O_2$ (mL) | Acetic acid (mL) | Num |
|---|---|---|---|---|---|
| 1 | 82.9071 | 4.09 | 30.11 | 53.66 | 1 |
| 1 | 82.8629 | 5.38 | 30.70 | 69.95 | 2 |
| 1 | 83.9883 | 4.07 | 30.43 | 57.53 | 3 |
| 1 | 82.4449 | 4.19 | 31.51 | 57.22 | 4 |
| 1 | 82.5795 | 4.86 | 30.91 | 63.83 | 5 |
| 1 | 82.8521 | 5.01 | 30.06 | 63.86 | 6 |
| 1 | 82.9873 | 4.02 | 34.54 | 66.15 | 7 |
| 1 | 82.6698 | 5.00 | 31.71 | 68.03 | 8 |
| 1 | 87.2345 | 4.00 | 30.18 | 65.66 | 9 |
| 1 | 82.6389 | 4.06 | 33.55 | 62.27 | 10 |
| 1 | 86.3181 | 4.14 | 30.26 | 64.85 | 11 |
| 1 | 83.4339 | 4.20 | 30.25 | 56.88 | 12 |
| 1 | 82.6921 | 5.50 | 30.25 | 69.06 | 13 |
| 1 | 82.3783 | 4.38 | 31.63 | 59.75 | 14 |
| 1 | 83.4678 | 5.26 | 30.60 | 69.91 | 15 |
| 1 | 82.8545 | 4.26 | 31.37 | 58.97 | 16 |
| 1 | 84.7008 | 4.19 | 32.84 | 68.50 | 17 |
| 1 | 83.1655 | 4.36 | 30.46 | 58.60 | 18 |
| 1 | 82.7543 | 5.12 | 31.06 | 67.70 | 19 |
| 1 | 83.2338 | 4.18 | 31.15 | 50.58 | 20 |
| 0.999 | 82.1909 | 4.00 | 30.00 | 50.48 | 21 |
| 0.900 | 78.5597 | 4.00 | 30.00 | 39.62 | 22 |
| 0.891 | 78.2138 | 4.00 | 40.52 | 70.00 | 23 |
| 0.809 | 75.1283 | 8.26 | 30.00 | 70.00 | 24 |
| 0.896 | 74.6328 | 4.00 | 30.00 | 28.26 | 25 |
| 0.596 | 67.1419 | 14.99 | 30.00 | 70.00 | 26 |
| 0.533 | 64.7711 | 4.00 | 57.06 | 20.00 | 27 |
| 0.243 | 53.9291 | 15.00 | 70.00 | 20.01 | 28 |
| 0.217 | 52.9417 | 15.00 | 70.00 | 36.75 | 29 |

In the triester spectrum, the peak in 1738 $cm^{-1}$ is related to carbonyl vibration and the absence of OH peak in 3400 $cm^{-1}$ verifies the triester structure. Furthermore in this spectrum, the peaks in 2923 $cm^{-1}$ and 2853 $cm^{-1}$ are related to stretching vibration of $CH_2$ and the peaks in 1465 $cm^{-1}$ and 1377 $cm^{-1}$ are related to bending vibration in $CH_2$ and $CH_3$. Also, the peaks in 1116 $cm^{-1,}$ 1170 $cm^{-1,}$ and 1245 $cm^{-1}$ are related to the stretching vibration of the ester C-O bond (Fig. 9). According to the FTIR spectrums, it can be concluded that in the epoxide spectrum, the disappearance of the peak in 844 $cm^{-1}$ is related to the opening of the oxirane ring (Fig. 10) and the formation of the peaks in 1706 $cm^{-1}$ and 1739 $cm^{-1}$ wavenumbers is associated with the formation of ester groups. Also, the peak in 3350 $cm^{-1}$ is related to OH group formation (Fig. 8). This peak has been disappeared in the final product (triester) (Fig. 9).

Table 7 shows the physicochemical properties of the final triester product and its comparison with ISO standard. As can be seen, the results for viscosity in both 40 and 100°C as well as the poor point are in line with the ISO VG-10 standard. The SWCO product however had higher VI and flash point than the standard which makes SWCO triester a better choice from the technical and safety point of view, respectively. It shows that the physicochemical properties of SWCO triester are compatible with those of the lubricants standard ISO VG10.

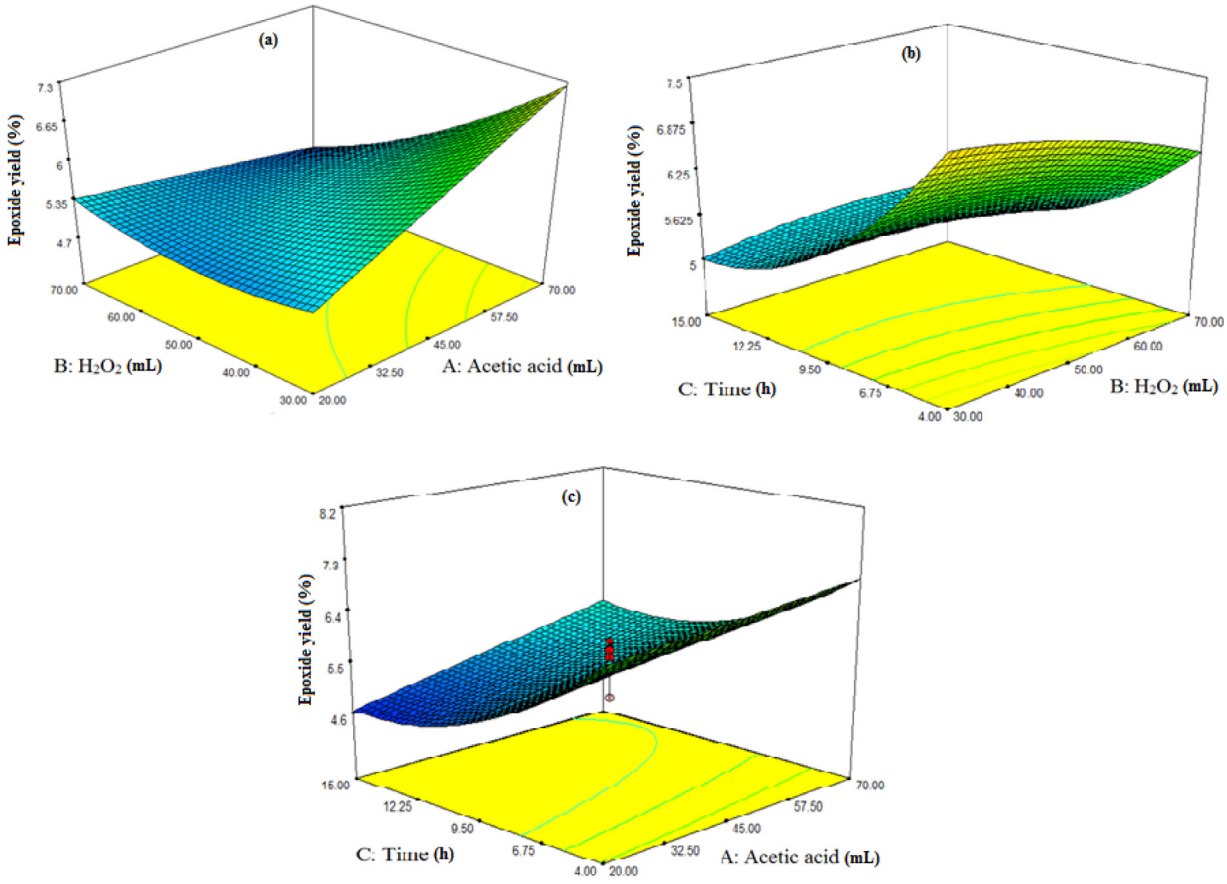



**Fig. 5.** Changes of epoxide yield (response) with variation in: (a) acetic acid and hydrogen peroxide; (b) time and $H_2O_2$; (c) time and acetic acid.

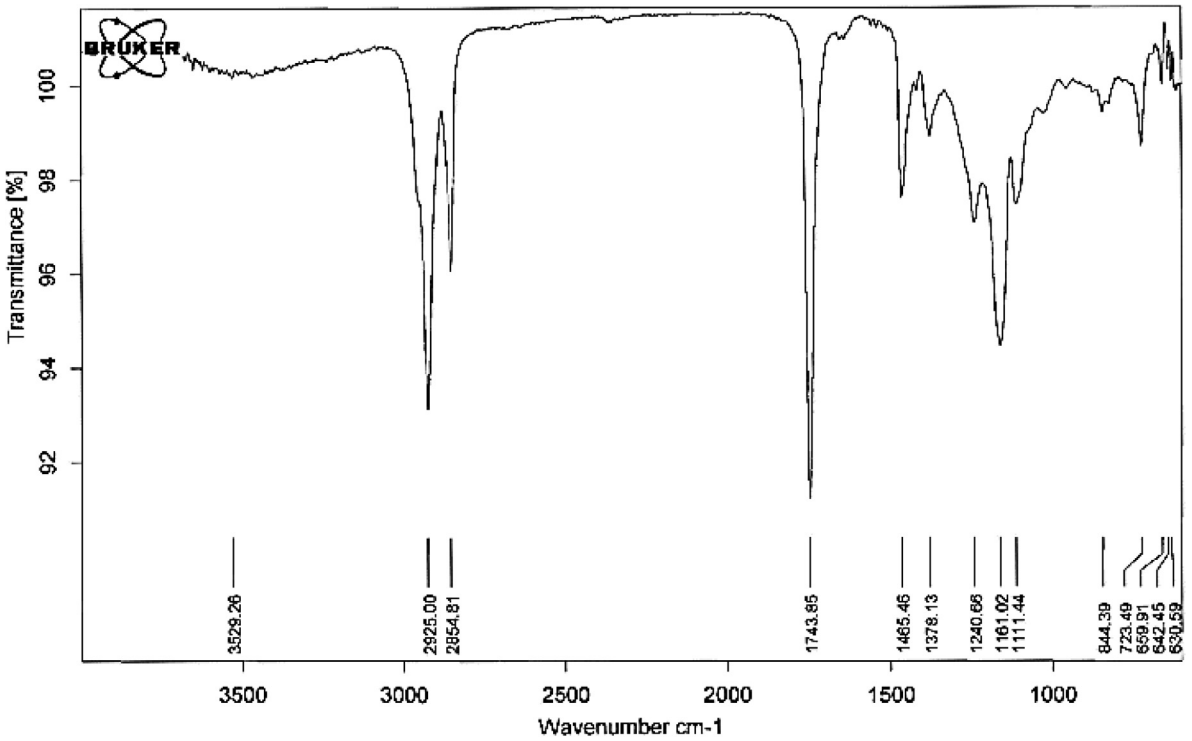



**Fig. 6.** FTIR spectrum of the epoxide.

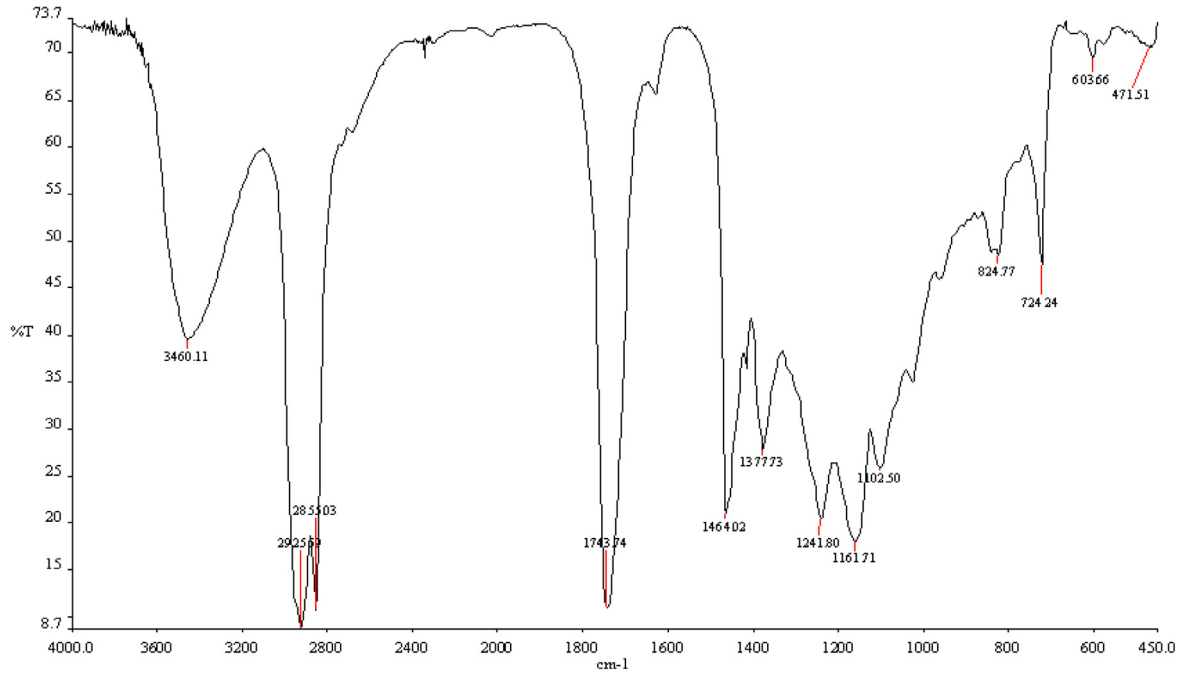



**Fig. 7.** FTIR spectrum of monoester.

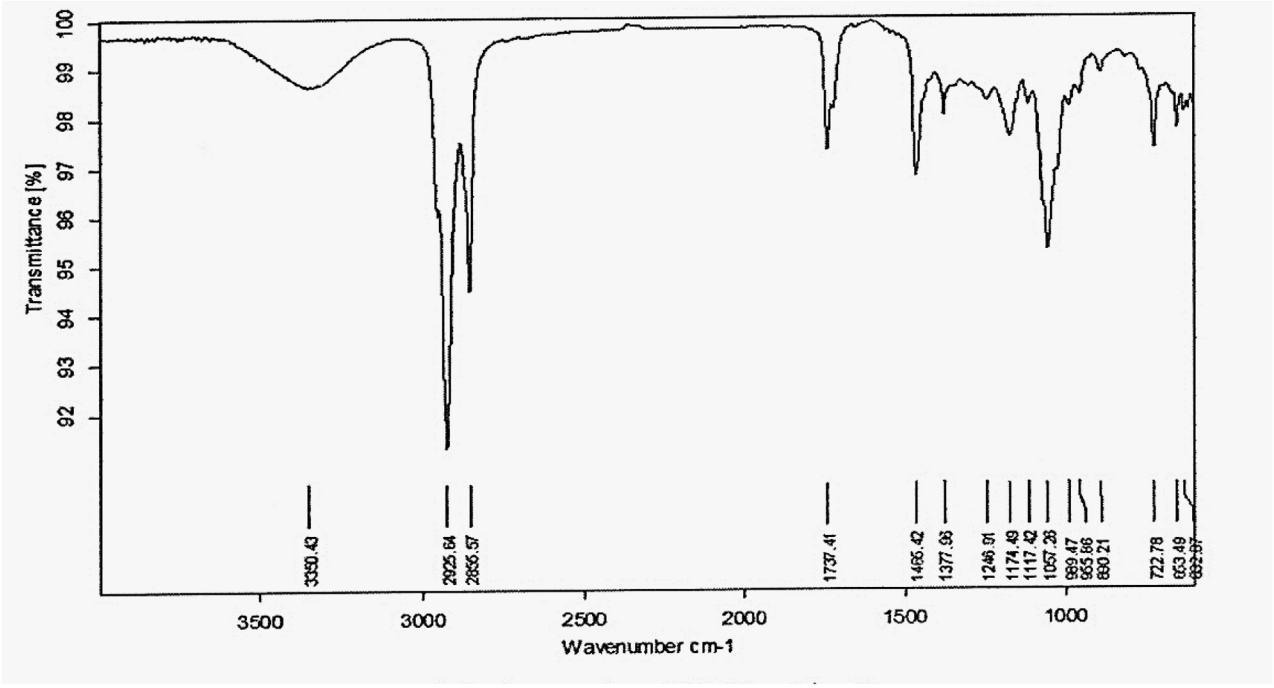



**Fig. 8.** FTIR spectrum of diester.

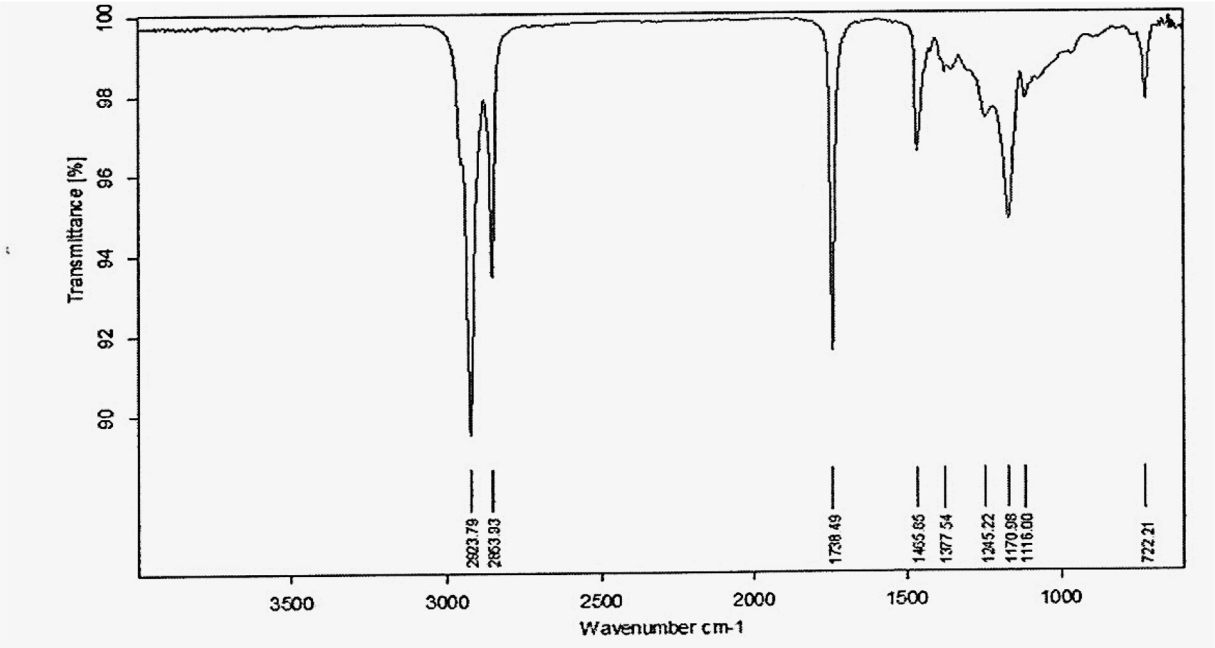



**Fig. 9.** FTIR spectrum of triester.

**Table 7**
Physicochemical properties of SWCO triester.

| Pour point (°C) | Flash point (°C) | Viscosity Index (VI) | Viscosity in 100°C (cSt) | Viscosity in 40°C (cSt) | |
|---|---|---|---|---|---|
| ASTM D97 | ASTM D93 | ASTM D2270 | ASTM D445 | ASTM D445 | ASTM standards |
| −30 | 235 | 135.12 | 3.57 | 11.91 | SWCO triester |
| −30 | 154 | 104 | >2.68 | >9 | ISO VG 10 |

## 4. Conclusions

Using RSM method, the optimum values of independent variables for having the most efficient reaction (82.91%) were obtained as 53.66 g of acetic acid, 30.11 g of hydrogen peroxide, and time of 4.09 h. The percentage of epoxide production reaction is in direct relationship with the increase of acetic acid and hydrogen peroxide and it has an inverse relationship with time. Also, it was found that the impact of acetic acid on efficiency was more than the impact of hydrogen peroxide. In the FTIR analysis of epoxide, the peaks in the area between 825 $cm^{-1}$ and 846 $cm^{-1}$ wavenumbers are indicative of oxirane ring entity (Fig. 8) which have been disappeared in monoester FTIR spectrum because of oxirane ring-opening (Fig. 9). Moreover, in the FTIR analysis of the triester, the disappearance of peaks in the area of wavenumber 3500 $cm^{-1}$ represents the triester formation. Furthermore, the peaks that are formed in the area between 1000 and 1200 wavenumbers indicate the formation of the COC ester bond. The physicochemical properties of the final product (triester) is compatible with ISO VG10 standard and can be applied as a biolubricant base oil.

## CRediT authorship contribution statement

**Shima Bashiri:** Writing - original draft, Investigation, Methodology, Software. **Barat Ghobadian:** Supervision. **Masoud Dehghani Soufi:** Data curation, Writing - review & editing. **Shiva Gorjian:** Writing - review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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